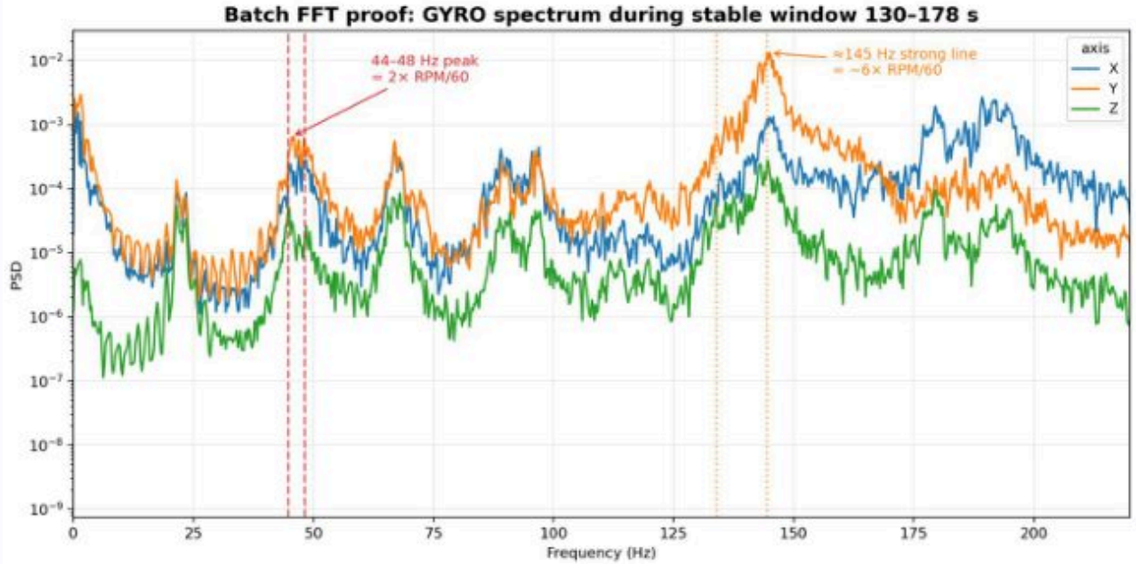
DARPA Lift Challenge | Gyro FFT cross-check

Cross-check the gyro spectrum before changing rate gains.

FFT Proof . Frequencies Present in the Log

BATCH gyro spectrum validates both filter targets before changing PID gains



Frequency interpretation

44–48 Hz line

Matches 2× RPM/60. Cover it with the RPM notch harmonics.

≈145 Hz dominant line

Strongest gyro peak; aligns with ≈6× RPM. Treat as a second narrow notch.

Why two notches?

One RPM notch follows changing motor speed; one throttle/reference notch targets the high-frequency line without making the RPM filter too wide.

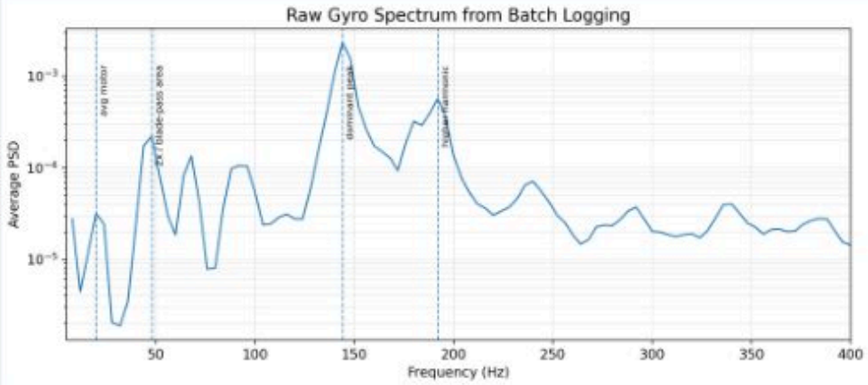
DARPA Lift Challenge | Gyro FFT cross-check

Cross-check the gyro spectrum before changing rate gains.

FFT Cross-check . Different Tools, Same Decision

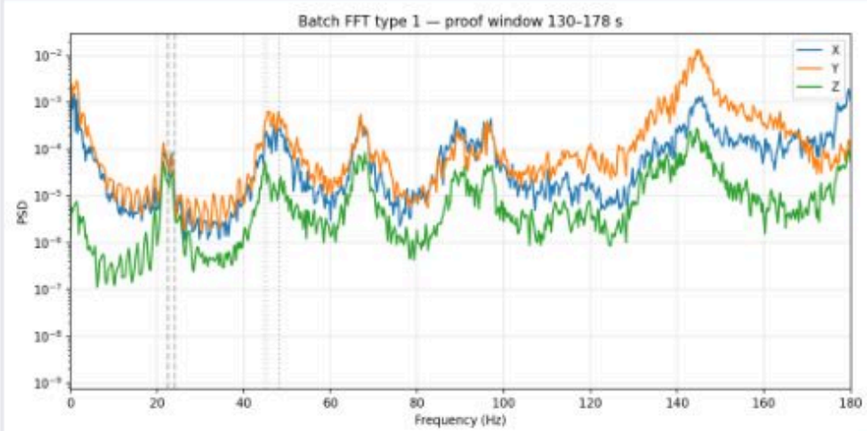
MAVExplorer-style spectrum and Filter Review/BATCH data point to the same targets

MAVExplorer / averaged gyro FFT



Dominant peak around ≈145 Hz; blade-pass area around ≈44–48 Hz.

ArduPilot Filter Review input (BATCH)



BATCH gyro data confirms the same frequency bands; use next flight to validate reduction.

Validation target after filter changes: lower energy near 45–48 Hz and 145 Hz without worsening RATE/PID tracking.